

Grade 1 Ontario Curriculum Algebra

Loop the Loop

Unit Overview: Loop the Loop

Grade: Grade 1 | **Strand:** Algebra

Target Expectations: C3.1, C3.2

Learning Goal: Children will write and execute simple code with sequential steps and loops, read existing code and predict outcomes, and alter code to change the result.

Lesson	Lesson Title	Learning Goal
Day 1	Step-by-Step Code	I can write step-by-step code for a simple task and follow it in order.
Day 2	The Loop Discovery	I can use a loop in my code to repeat steps.
Day 3	Reading Code	I can read code and predict what will happen when it runs.
Day 4	Altering Code	I can change code and predict the new outcome.
Day 5	My Code Capstone	I can write my own code to do a task with sequence and loop.

Lesson 1: Step-by-Step Code

Goal: Code is a list of instructions in ORDER.

Expectation: C3.1

Learning Goal: "I can write step-by-step code for a simple task and follow it in order."

Materials: M1_grid_mat, M2_code_cards, M4_robot_marker, M10_loop_poster

1. Minds On (12 mins): Coda's Sandwich Code

How do you make a peanut-butter sandwich? Tell Coda STEP BY STEP. Coda opens with the unit refrain: 'STEP, by STEP, by STEP — that's CODE!' (Class echoes.)

Bring children to the carpet. Reveal Coda the Coding Robot. 'Friends — meet Coda. Coda is a CODING ROBOT. Coda only does what we tell it — STEP by STEP. Today we'll write CODE for Coda.'

2. Action (25 mins): Code-and-Execute Stations

- Whole group: code Coda's path on the grid.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station B: include a path that requires turning (e.g., from (0,0) to (3,3)).
- Discuss order matters.
- Circulate to record formative evidence on the sequence tracker.
- STRATEGY COMPARE: 2 pairs share their step-by-step solutions for the same path.

3. Consolidation (13 mins): Coda's Sandwich Brag

Discuss:

- What is CODE?
- Why does the ORDER of steps matter?
- What happens if you swap two steps?
- Can you give an example of code from your day?
- Coda's preview: tomorrow Looper will show us how to SHORTEN our code with LOOPS (Lesson 2 — C3.1).

Exit: Coda's refrain — class chants three times together: "Coda says: STEP, by STEP, by STEP — that's CODE!" Each child writes ONE Coda command on a sticky and tags it with an action verb.

Lesson 2: The Loop Discovery

Goal: When we repeat the same step many times, we can use a LOOP.

Expectation: C3.1

Learning Goal: "I can use a loop in my code to repeat steps."

Materials: M1_grid_mat, M2_code_cards, M4_robot_marker, M5_loop_card

1. Minds On (12 mins): Looper Comes to the Rescue

Coda needs to take 5 steps right. Watch what Looper does. Looper opens: 'Pick the BIG number!' Class chants: 'Loops make it SHORT!'

Bring children to the carpet. Coda + Looper puppets. 'Friends — yesterday Coda followed step-by-step code. Today Coda's friend Looper has a TRICK that makes code shorter.'

2. Action (25 mins): Loop Stations

- Whole group: practice 4 more loop conversions.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station B: include a hexagon (Repeat 6: forward, turn).
- Discuss the why.
- Circulate to record formative evidence on the loop tracker.
- STRATEGY COMPARE: 2-3 pairs share alternative loop solutions for the same shape.

3. Consolidation (13 mins): Looper's Brag

Discuss:

- What is a LOOP?
- When should you use a loop?
- Why are loops better than writing the same step many times?
- What does 'Repeat 4: forward, turn' draw?
- Tomorrow we'll READ Coda's code on the grid (Lesson 3 — C3.1).

Exit: Looper's call-and-response refrain — Looper calls 'Pick the BIG number!' and class chants back 'Loops make it SHORT!' Repeat 3x.

Lesson 3: Reading Code

Goal: We can READ code and predict what it will do BEFORE running it. Predicting tests our understanding.

Expectation: C3.2

Learning Goal: "I can read code and predict what will happen when it runs."

Materials: M1_grid_mat, M3_existing_code_cards, M4_robot_marker, M7_predict_chart

1. Minds On (12 mins): Coda's Code Mystery

Coda gives you code BEFORE running. Can you PREDICT where Coda will end up? Coda opens: 'PREDICT then CHECK — that's how we read code!'

Day 1 we WROTE step-by-step code with Coda. Day 2 Looper showed us LOOPS. Today we READ code Coda hands us. Tomorrow we'll ALTER it.

2. Action (25 mins): Predict-and-Check Stations

- Whole group: predict outcomes for 3 more codes.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: include 2 codes with NESTED loops or longer sequences.
- Discuss when predictions go wrong.
- Circulate to record formative evidence on the prediction tracker.
- STRATEGY COMPARE: 'How did YOU predict Coda's destination?' Some children traced on paper; some pictured Coda walking; some used a finger on the grid.

3. Consolidation (13 mins): Coda's Predict Brag

Discuss:

- What does it mean to PREDICT?
- How do you predict where Coda ends up?
- What happens when your prediction is wrong?
- How does predicting help you understand code?
- Tomorrow we'll ALTER Coda's code (Lesson 4 — C3.2).

Exit: Coda's refrain — class chants together: "PREDICT then CHECK — that's how we read code!" Then call-and-response with Looper: 'Loops make it SHORT!' Each child writes their PREDICT square on a sticky...

Lesson 4: Altering Code

Goal: When we change code, the outcome changes.

Expectation: C3.2

Learning Goal: "I can change code and predict the new outcome."

Materials: M1_grid_mat, M3_existing_code_cards, M4_robot_marker, M7_predict_chart

1. Minds On (12 mins): Coda's Code Edit

Original code: Repeat 3 right. Now we change 3 to 5. What happens?

Bring children to the carpet. Coda + Looper. 'Friends — yesterday we PREDICTED what code does. Today we CHANGE code and predict the NEW outcome.'

2. Action (25 mins): Alteration Stations

- Whole group: alter 4 more codes.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: invent an alteration that DOESN'T change the outcome.
- Discuss the alteration framework.
- Circulate to record formative evidence on the alteration tracker.

3. Consolidation (13 mins): Coda's Edit Brag

Discuss:

- What does ALTER mean?
- How does changing a number in a loop affect the outcome?
- How does swapping steps affect the outcome?
- Are there changes that DON'T affect the outcome?

Exit: Each child writes ONE alteration on a sticky note: 'Original: ____.

Lesson 5: My Code Capstone

Goal: We can write our own code to do a task — drawing a shape, walking a path, telling a story.

Expectation: C3.1, C3.2

Learning Goal: "I can write my own code to do a task with sequence and loop."

Materials: M1_grid_mat, M2_code_cards, M5_loop_card, M9_capstone_template

1. Minds On (12 mins): Coda's Project Showcase

Today YOU write code. Pick a project: draw a shape, walk a path, tell a tiny code story. Then we share!

Bring children to the carpet. Coda + Looper. 'Friends — today is YOUR day. We've learned sequence (Lesson 1), loops (Lesson 2), reading code (Lesson 3), altering code (Lesson 4). Now you put it ALL together.'

2. Action (25 mins): Capstone Code Stations

- Hand out capstone templates.
- Children work for 10-12 minutes.
- Stretch challenge: include 2 loops in the code.
- Gallery walk: pairs swap templates and execute each other's code on the grid.
- Whole-class share: 5-6 children share their project aloud.

3. Consolidation (13 mins): Closing Brag and Certificate Ceremony

Discuss:

- What's the most creative project you saw today?
- How did your loop make your code shorter?
- What's something about coding you'll remember?
- Where might you use coding in real life?

Exit: Each child receives a Junior Coder certificate.

Date: _____

Learning Goal: I can write step-by-step code for a simple task and follow it in order.

1. Write 5 steps in order for a simple task (brush teeth, walk to school, get dressed).
2. Each grid shows Coda starting at the bottom-left.
3. Invent your own simple task.

[illegible]

Name: _____

Date: _____

Looper's Loop Workshop

Learning Goal: I can use a loop in my code to repeat steps.

Instructions for the Student:

1. Each long code can be written shorter using a loop.
2. Write loop code to draw each shape.
3. Invent your own loop.

Part 1: Long Code to Loop Code

Each long code can be written shorter using a loop. Write the loop version.

[illegible]

Name: _____

Date: _____

Coda's Predict Workshop

Learning Goal: I can read code and predict what will happen when it runs.

Instructions for the Student:

1. Read each code.
2. Read each code.
3. Read this code with a loop.

Part 3: Loop Code Predict

Read this code with a loop. Predict where Coda ends.

[illegible]

Date: _____

Learning Goal: I can change code and predict the new outcome.

1. Each code is altered by changing a number.
2. Each code is altered by changing a step.
3. Find an alteration that DOESN'T change the outcome.

Each code is altered by changing a step. Predict the new outcome.

[illegible]

Name: _____ Date: _____

Name: _____ Date: _____

My Code Project

Learning Goal: I can write my own code to do a task with sequence and loop.

Instructions for the Student:

1. Project name and 1-sentence description.
2. Write 4-8 steps of code.
3. Draw or describe what your code does.

Part 1: My Project

Project name and 1-sentence description.

[illegible]

Coding Grid Mat

(M1_grid_mat)

Letter-size 5x5 grid mat for code execution.

Robot Grid Mat

	0	1	2	3	4	5	6	7
0								
1								
2								
3								
4								
5								

PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 mat per pair.

2. Laminate.

3. Hand out at start of Lesson 1.

4. Use through Lesson 5.

5. Optional: extras.

USE IN CLASS

Quantity: 1 mat per pair

Lessons: 1, 2, 3, 4, 5

Prep: ~15 min

Code Step Cards

(M2_code_cards)

30 cards: step left/right/up/down, repeat N, etc.

Code Cards

MOVE 1 RIGHT	MOVE 1 LEFT	MOVE 1 UP	MOVE 1 DOWN
TURN LEFT	TURN RIGHT	REPEAT 3	STOP

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 5

Laminate: yes

USE IN CLASS

Quantity: 30 cards

Lessons: 1, 2, 3, 4, 5

Prep: ~25 min

TEACHER PREP

1. Print 5 pages on cardstock.

2. Laminate.

3. Cut into 30 cards.

4. Sort by type.

5. Distribute Lessons 1-5.

Existing Code Sample Cards

(M3_existing_code_cards)

12 pre-written code samples for reading and predicting.

Sample Cards

MOVE 3	TURN	REPEAT 2
STOP	MOVE 5	TURN LEFT

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 12 cards

Lessons: 3, 4

Prep: ~12 min

TEACHER PREP

1. Print 2 pages on cardstock.

2. Laminate.

3. Cut into 12 cards.

4. Sort by difficulty.

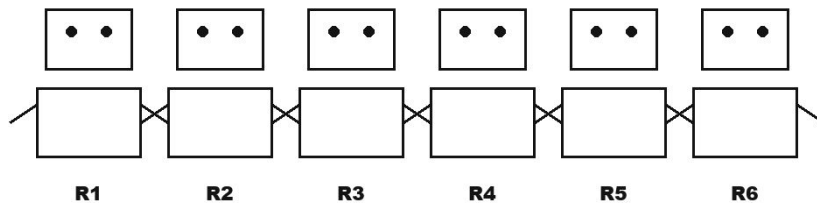
5. Distribute Lessons 3-4.

Robot Marker

(M4_robot_marker)

Small token (paper robot or dice) that moves on the grid.

Robot Tokens



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 10 tokens

Lessons: 1, 2, 3, 4, 5

Prep: ~8 min

TEACHER PREP

1. Print 1 page.

2. Cut into 10 tokens.

3. Sort into pair sets.

4. Have ready Lessons 1-5.

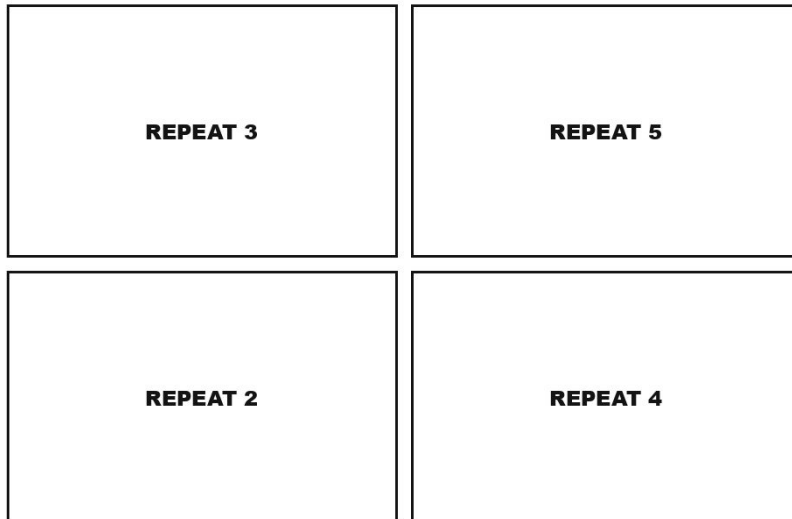
5. Optional: dice instead.

Loop Cards

(M5_loop_card)

8 cards with loop counts: Repeat 2, 3, 4, 5.

Loop Cards



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 8 cards

Lessons: 2, 5

Prep: ~10 min

TEACHER PREP

1. Print 1 page on cardstock.
2. Laminate.
3. Cut into 8 cards.
4. Distribute Lessons 2 and 5.
5. Optional: extras.

Code Writing Template

(M6_code_template)

Letter-size template for writing custom code.

Template

Code template — write your code:



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

USE IN CLASS

Quantity: 1 per child

Lessons: 5

Prep: ~12 min

TEACHER PREP

1. Print 1 per child.
2. Hand out Lesson 5.
3. Children fill in.
4. Collect for assessment.
5. Display gallery.

Predict-and-Check Chart

(M7_predict_chart)

Letter-size chart with PREDICT and CHECK columns.

Equation Table

Step	Predict	Actual

PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS

Quantity: 1 per pair
Lessons: 3, 4
Prep: ~12 min

TEACHER PREP

1. Print 1 chart per pair.
2. Laminate.
3. Hand out Lessons 3-4.
4. Use dry-erase.
5. Reuse.

Code Alteration Cards

(M8_alter_cards)

8 cards each with original code + a change.

Alter Cards

swap two steps	add a STOP
remove a step	add a TURN
double the loop	halve the loop

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 8 cards

Lessons: 4

Prep: ~10 min

TEACHER PREP

1. Print 1 page on cardstock.

2. Laminate.

3. Cut into 8 cards.

4. Distribute Lesson 4.

5. Optional: extras.

Capstone Code Template

(M9_capstone_template)

Letter-size template for Lesson 5 capstone.

1. The PROJECT:
2. The CODE:
3. The LOOP:
4. The OUTCOME:

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

USE IN CLASS

Quantity: 1 per child

Lessons: 5

Prep: ~12 min

TEACHER PREP

1. Print 1 per child.

2. Hand out Lesson 5.

3. Children fill in.

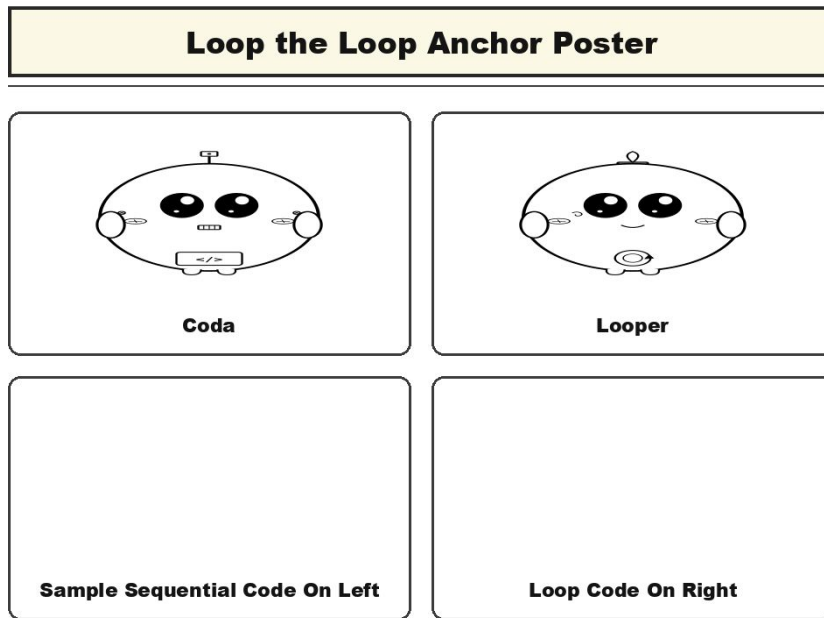
4. Collect for assessment.

5. Display gallery.

Loop the Loop Anchor Poster

(M10_loop_poster)

Large classroom anchor showing Coda and Looper with sample code.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

USE IN CLASS

Quantity: 1 poster

Lessons: 1, 2, 3, 4, 5

Prep: ~18 min

TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall Day 1.

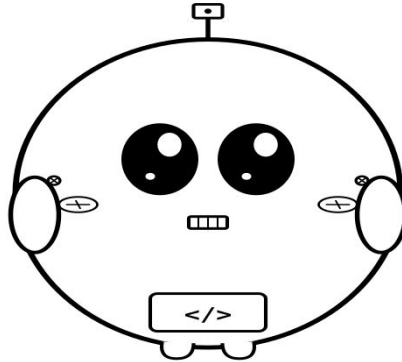
4. Reference all week.

5. Roll for re-use.

Coda the Coding Robot

(char_coda_puppet)

Printable Coda illustration on a popsicle stick.



Coda the Coding Robot

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

4. Optional: laminate.

5. Store flat.

USE IN CLASS

Quantity: 1 puppet

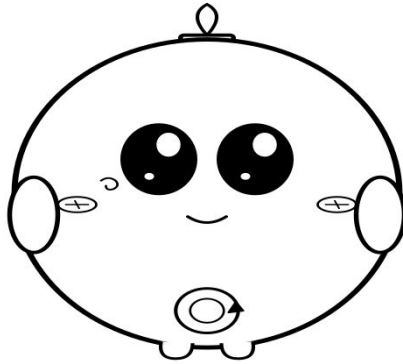
Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

Looper the Loop Helper

(char_looper_puppet)

Printable Looper illustration on a popsicle stick.



Looper the Loop Helper

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 2, 5

Prep: ~10 min

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

4. Optional: laminate.

5. Store flat.

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
C3.1 solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events	Cannot order steps in code.	Writes 3-step ordered code with prompts.	Writes 5+ step ordered code with at least one loop.	Writes complex code with multiple loops; executes correctly.
C3.2 read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes	Cannot read code or predict outcomes.	Predicts simple sequential outcomes with prompts.	Predicts outcomes for code with loops; alters code and predicts the new outcome.	Articulates how alterations affect outcomes; identifies invariant alterations.

Junior Coder Certificate

This certificate is proudly presented to:

For successfully completing the 5-Day Loop the Loop unit and demonstrating the skills of a Junior Coder. Coda and Looper congratulate the student.

By completing this unit, this student has demonstrated:

- Writing step-by-step code in order
- Using loops to compress repetition
- Reading code and predicting outcomes
- Altering code and predicting new outcomes
- Inventing a coding project with sequence and loop

Marketplace Listing

A 5-day Ontario Grade-1 unit on coding without screens (C3.1-C3.2) framed around 'Loop the Loop' — Coda the Coding Robot and Looper the Loop Helper.

Price: \$9.99 CAD

Pages: ~32

Classroom time: 250 min

Teacher prep: ~90 min

Top tags: grade-1, math, ontario-curriculum, coding, algebra, without-screens, canadian

What's included:

- 5 fully-scripted lesson plans with 50-minute durations and 250 minutes of total teaching time
- 5 student worksheets aligned to each lesson
- 1 mid-unit formative assessment (Code Check-In)
- 10 printable teacher manipulatives plus 2 character puppets: coding grid mat, code step cards (30), existing code sample cards (12), robot marker template,...
- 1 end-of-unit reflection sheet with 5 self-assessment prompts
- 4 class observation trackers: 1 diagnostic + 3 formative
- 1 two-row summative rubric covering C3.1-C3.2 across 4 levels
- 1 summative-administration guide tying the rubric to Lesson 5 worksheet
- Junior Coder certificate (one per child)
- Detailed expanded walkthroughs covering common misconceptions
- Differentiation tracks built into every lesson